

11.07.26

CO1 AT-1 SHORT ANSWER TEST.

1. What is Data Science? Explain its Primary Objective.

Ans: Data Science is an interdisciplinary field that uses Scientific Methods, Statistics, Programming and Machine Learning to Extract Meaningful insights from data.

PRIMARY OBJECTIVE:

- To Collect, Process, analyze, and interpret Data.
- To Support better Decision-making and Solve Real World Problems.

2. Why is Data Science Important in today's Real world? Mention Any 4 Reasons.

Ans: Data Science is Important because it:

- Helps organizations to Make data-driven decisions.
- Predicts future trends using historical data.
- Improves business efficiency / Productivity.
- Enables Personalized Customer exp.

3. List Any 4 Benefits of Data Science in Business and Society.

Ans:

- Better Decision Making
- Increased Business Profits
- Improved Healthcare & Treatment
- Fraud Detection and Enhanced Security

4. Mention 4 Real World Applications of Data Science

- Healthcare (Disease Prediction)
- Banking (Fraud Detection)
- E-Commerce (Product Recommendation)
- Transportation (Traffic & Route)

5. What are the Different facets of Data? Explain each.

- STRUCTURE - Organization of Data (Structured, Semi-Structured, Unstructured)
- SOURCE - Origin of Data (Sensors)
- FORMAT - Form in which Data is Stored (CSV, JSON, XML)
- QUALITY - Accuracy, Completeness, Consistency & Validity of Data
- VOLUME - Amount of Data Collected.

• VELOCITY - Speed at which Data is generated & Processed.

• VARIETY - Different forms of Data such as Text, Images.

• VALUE - Usefulness of Data in Making Decisions

6. Differentiate b/w Structured, Unstructured & Semi-Structured w' examples

TYPE	DESCRIPTION	EXAMPLE
STRUCTURED	Organized in rows and columns.	SQL Database
UNSTRUCTURED	Has tags but no fixed table.	XML
SEMI-STRUCTURED	No predefined format	IMAGES

7. What is Big Data? State its Key Characters (5V's)

Big Data Refers to Extremely Large & Complex Datasets that cannot be processed using traditional methods

5V's

- Volume
- Velocity
- Variety
- Veracity
- Value

8. Explain Big Data Ecosystem.
• Name 4 Technologies.

The Big Data Ecosystem is a Collection of tools & Technologies used to Collect, Store, Process & Analyse Large Datasets.

- Hadoop
- Apache Spark
- Apache Hive
- Apache Sqoop

9. What are the Major Phases of Data Science?

- Data Collection
- Data Cleaning
- Data Integration
- Data Transformation
- Data Analysis
- Data Visualization
- Model Building
- Decision Making

10. What is Data Retrieval?
Mention two common Sources

Data Retrieval is the Process of Obtaining Data from Storage.

- Databases
- Websites / API's

11. Data Collection Importance

- It Provides Accurate & ~~Real~~ Info
- Improves Analysis Quality
- It Supports Better Decisions
- Reliable Decision Making

12. What is Data Cleaning? 4 Techniques

Data Cleaning is the Process of Identifying and Correcting Wrongy Data

TECHNIQUES:-

- Remove Duplicate Records
- Handle Missing Values
- Remove Outliers
- Correct Wrong Data

13. Data Integration

Process of Combining Data from Different Sources

- Improves Data Quality
- Eliminates Redundancy
- Provides full View
- Accurate Analysis

14. Data Transformation

Data Transformation is the Process of converting Data to suitable format for analysis

TECHNIQUES

- Convert text to Numeric
- Normalizing Data
- Changing Data forms

15. Data Analysis

Data Analysis is the Process of Examining & Interpreting Useful Information from Data

- Statistical Analysis
- Clustering

90%

$$\frac{28}{30}$$

$$\frac{11}{11} \times \frac{4}{4}$$